

HCCDA AI Course Week 1 Day 2 Assignment

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1. What is NLP?

Natural Language Processing (NLP) is a part of Artificial Intelligence that helps computers understand human language. It works with text and speech and helps computers understand what people are saying or writing. For example, when we ask a question from a chatbot, NLP helps the computer understand our question and give a suitable answer. NLP is also used in voice assistants, search engines, and translation apps.

2. Difference between NLP and Traditional Text Processing

Traditional text processing is used for simple tasks like finding a word, counting words, removing spaces, or changing text into lowercase. It usually works with fixed rules. NLP is more advanced because it tries to understand the meaning of the text. For example, traditional text processing can find the word “good” in a sentence, while NLP can understand that the whole sentence is showing a positive opinion.

3. Common Tasks in NLP

There are many tasks in NLP. Tokenization means breaking text into smaller parts, usually words or sentences. For example, “I like Python” can be divided into “I”, “like”, and “Python”. Lemmatization changes words into their basic form. For example, “running” and “runs” can be changed to the basic word “run”. Sentiment Analysis is used to find whether a text has a positive, negative, or neutral opinion. For example, “This phone is great” has a positive sentiment.

4. Applications of NLP

NLP is used in many applications that we use in daily life. Chatbots use NLP to understand questions and reply to users. Translators use NLP to translate text from one language to another. Email filters use NLP to detect spam emails by checking their content. NLP is also used in search engines, voice assistants, grammar checkers, and other applications.

5. Role of Deep Learning in NLP

Deep Learning has made NLP systems better and more powerful. It allows computers to learn from a large amount of text and find patterns in it. Deep Learning models can be used for translation, sentiment analysis, question answering, and text generation. Models such as RNNs, LSTMs, and Transformers are commonly used in NLP. Transformers are especially useful because they can understand the relationship between words in a sentence. This helps modern NLP systems give better and more accurate results.